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Colleges aantekeningen

History of Science (Vrije Universiteit Amsterdam)

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Week 1, college 1

- Earlier, only the rich and people like kings had access to machines.
- Changed in the early 19th century when the whole western Europe was confronted with the steam engine.
    - It was discovered that engines could serve purposes.

Babbage

- He was a mathematician interested in how to make economy efficient with machines and manufactures.
- He wrote a book *on the economy of machines and manufactures*
    - In which he discussed several business that he had visited and how these could have been made more efficient.
    - This he did mostly by spotting the bottleneck in the process by much analyzing.
        - And thinking of a solution.
- He came up with the process of the clearing house.
    - Here the bankers of London would meet and exchange checks every week.
    - Babbage came up with the idea of every bank opening an account with the bank of England.
        - Thereby reducing the whole operation to an accounting trick.
        - It still took the banks a few decades to implement the system.

Frederick Taylor

- He came up with a more systematic approach than Babbage had been doing.
    - He came with principles that would allow manufactures to get the bottleneck themselves and prevent it.
        - But also a much further division of labour than Babbage had imagined.
        - He inspired for example Ford.
- One of the people contributing to the efficiency movement was Henri Fayol.
    - Found out that it would be more convenient if the French bureaucracy would be adapted to similar processes as well.
        - His efficiency was more focussed on the administration part.
    - He came up with solutions to bottlenecks in the administrative part of business.
- Taylor and Fayol and the efficiency movement were part of a movement were people would put more trust in machines but only in numbers.
    - Trust in numbers was a huge benefit to governments and industrials.

- Taylor came with principles that would allow manufacturers to get the bottlenecks themselves and try to make their work around.
    - He came with principles that would allow manufacturers to get the bottlenecks themselves and try to find their work around.
    - He also came with a much further division of labour.
    - This efficiency not only by using machines but also efficiency by implementing a division of labour.
    - This efficiency was a huge movement in industrials and manufacturers.
- Henri Fayol was an European that was introduced to Taylor's work.

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    - Introduced this in some of the mining companies in France.
    - He came up that the French bureaucracy should do same principles as well.
    - His efficiency is much more focussed on the administration part.
        - Instead of the industrial part that was covered by Taylor.
    - He comes up with various solutions to the bottlenecks in the administrative parts in business.

Hollerith

- In the late 19th century, Hollerith came up with a machine that could speed up the US census.
    - The 1890 census was the first mechanised process were machines counted.
    - The machines did it even much quicker and people in higher circles were quickly convinced about the machines.
        - Eventually the trust in numbers also slides down to the normal people.

- European continent was politically more “leftish” than US/UK.
    - The public was not yet convinced of statistic and machines.
    - The offices and other things were already existing for decades and centuries and already had their ways of doing procedures.
        - Introducing new ways or machines was not something they thought about.
        - Tradition can hold someone back also which you can see here.
- The cash register met opposition in Europe whereas in the US it was embraced quickly.
- So the trust in numbers and machines were rising in the early 20th century.
- The idea that machines could make life easier was in the heads of industrialist and governments in the early 20th century.
    - And it would increase a more important wish.
        - An ideal that people strove for.
- It was for 3 traditions that people strove for automation:
    - Administration
    - Process control
    - Science
- In the US the focus was on the administration control.
    - In Europe the focus was more on the other 2.
    - But all 3 traditions were present on both continents.

Administration tradition

- Around the 1870s, clerks were still copying letters by hand.
- But around 1920, the typical office was full of typewriters to copy letters.
    - You could make up to 10 copies at once.
    - Administration of companies was changed completely in a few decades.
- In administration punched card machines were introduced.
    - These were used for example for sorting, counting and tabulating.
- You also had calculators and typewriters.
    - All sorts of machines were introduced in the administrative business.

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- The efficiency movement struck Europe and you had all kinds of international exhibitions which sold these kinds of machines.
- Administrations were enthusiastic about these kinds of machines.
    - And tried to introduced it in their business with things like Fayol and Taylor their principles.
- the company ROBAVER introduced machines in 1922.
    - This was a huge success.
- PCGD introduced machines in 1923.
    - Was a complete failure.
    - Machines were introduced in a company that was in a transit.
        - People who had an idea of what the machines could do and how were fired in the transition.
- The Dutch State Mines introduced in 1923.
    - Was a success.
- So in some companies the machines were introduced but in some they were not.

Efficiency movement:

- 1923
    - Administratieve arbeid, maandblad voor rationele werkmethoden
- 1925
    - NIVE, a study group for administrative appliances.
- 1927
    - Studiekring moderne kantoortechniek.
- 1935
    - Kantoormachine Gids.
- 1938
    - Tijdschrift voor organisatie en efficiency.

- Not necessarily all of these efficiency movements things were high tech.
- All these administration efficiency reaches try to achieve efficiency by following a business outline.
    - A diagram for example.

Process control

- You can think of large oil refinery installations or steel production.
- These processes you don't want to be close to.
    - High temperatures, low amounts of oxygen etc.
    - You want to keep an eye at a distance.
- These kinds of traditions were automated as well.
    - This was a different kind of automation.
    - Efficiency was involved but this was a complete different kind.
- There were machines used which you could call computers.
    - That was a different kind of automation than administrative business.
        - The machine here took the initiative to intervene.
            - They were still checked.
        - In administration business it was always a person making the decision.
- In the process control there were outlines of the process like:

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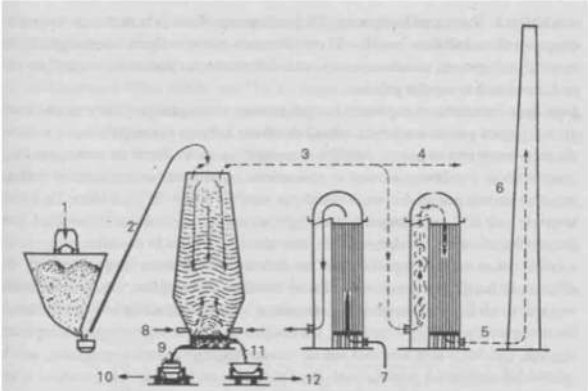

- The machines controlled these kinds of processes.
    - All decisions it could make and responses could be drawn in an outline.

Science and engineering.

- It were computers (people) that were active in here.
    - In all kinds of weather prediction, aeronautics etc.
- These were first analogue computers/machines like the differential analyser.
    - That were able to perform one specific type of calculation.
    - Had to be specified if they had to do another calculation.

- These 3 traditions were all trying in their way to achieve automation.
    - they all had their ideas of what a machine should do:
        - Administration:
            - That was not necessarily a high tech machine but to achieve efficiency.
            - The felt urgency to achieve automation was highest.
            - The administration business had most money to put in.
                - So from this business the actual first computer came in.
        - Process control:
            - That was analogue technology that was playing a growing role in the control of processes.
            - The technology was already there.
        - Science:
            - It was working on the various models that were being produced in the 20th century.
            - The concepts and calculations were being performed.
        - All 3 of them had outlines in common which they used to define the process.

- It was from 3 traditions were people strove for automation.
    - Administration
    - Process control
    - Science
- In offices were administration happened. firstly there were a lot of clerks doing copying by hand.

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    - After that the typewriter fully did copying which could do more than 1 at the same type.
    - The administration of larger companies was changing.
    - Also punch card machines were introduced in administration.
        - Punched card typing
        - Calculators
        - Typewriters
        - Sorting machines
        - Counting machines
        - Tabulating machines.
    - All these administration gadgets were not necessarily very high tech.
    - In administration business outlines were drawn.
- In process control we can think of large oil refinements or steel producers.
    - People don't want to be near these processes because of temperatures.
        - So people needed to keep an eye on a large distance.
    - These installations were becoming automated as well.
        - Different kind of efficiency and automation.
    - There were machines produced there which at eyesight you can call computers.
        - There were feedback loops which could take appropriate action which took over an intervention of a human being.
        - Here the machine took appropriate action itself but in administration it was always a person that makes the decision.
            - Machines were checked in process control.
    - Also outlines were made.
- In science and engineering were computers which first were humans and who did calculations for:
    - Weather prediction.
    - Mechanical calculations etc.
    - This 3th tradition would involve scientist and engineers doing these calculations or designing calculations for people to do them.
    - First there were analogue computers which were able to perform one specific type of calculation.
        - Needed to be adjusted to do another type of calculation.
    - Most calculations were done on worksheets.
- These 3 traditions were all on their way striving to automation.
    - They all had their ideals about what a machine would do.
    - For administration that was not necessarily a high tech machine.
        - It was simply a way of achieving efficiency.
    - For process control that was analogue technology that played a growing role in the control of processes.
    - In science it was working on various model that were being produced.
    - All 3 of them had outlines in common.
        - They used outlines to define the processes.
    - From the administration business rose the first computer because they had the most money to put in.

Week 2, college 1

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- Britain claimed other than the US not so devastated by the war.
    - So they could start a serious computer research program.
- In Japan in 1944-1948 there was the Yamashita statistical machine.
    - Which was claimed by the government because before the war they relied on IBM machines.
        - To build a new statistical machine at the beginning of the war became crucial.
        - It had an actual typewriter to add the data.

Background

- For the US and UK the machines were not interested as an statistical project.
    - Because they had IBM machines.
- They wanted machines for Manhattan project, code breaking, radar etc.
    - IBM machines would provide a good job, but they were not particularly build for it.
    - Scientist tried to provide a machine that was faster, more reliable.
- After WWII the need for scientific calculations exploded.
    - With the rise of NASA and atomic power.
    - At NASA for quite some time, people were still computers.
        - But computers we know today took slowly over.
- These progresses were also made in continental Europe of the late 1940s as well.
    - It was devastated, but the Marshall plan and determination to rebuild the nation speeded things up.
    - Building a computer was not relatively expensive.
    - Within Europe there was a mix of fatalism and optimism.
        - Fatalism in the sense that Europe was shot to pieces.
        - Optimism in the sense that people were working to achieve a better society.

Cold war science

- Early examples of the need of scientist and them being helpful in rebuilding the nation was the 1946 of the mathematical centre here in Amsterdam.
- Within Europe there were people involved in the Moore school.
- Hans Freudenthal claimed that calculators could win the war.
    - It was reflecting that mathematicians contributed to the victory in the war.
- The question was if someone could build a machine that could do all kinds of calculations.
- Cold war Europe was not following the US in their machines.
    - They thought how to compete with stuff coming in from the US to Europe.
    - A strong need was felt to create their own tech, representatives, unions etc.
        - Computers was part of it as well.
    - Within every European country there were scientist trying to build a machine.
    - They did not want to be completely dependent on US culture.
- During the war many machines were in progress to help in war effort.
    - Because the war ended earlier than expected, the parts were sold cheap.
    - Almost all countries began developing computer machines.
        - They were there to prove that computers were not something that belonged to persons.

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        - They were built for nationalistic things.
- Were the Japanese relied at electronic storage of knowledge, the machines in the UK were used for calculations, so they weren't stored.
    - The programs were stored.
    - Number storage was not needed.
        - You wanted the results printed.
    - One of the ways to carry the bit to another thing you were counted you needed a delay.
        - That was achieved with a Mercury delay line.

Dutch dinosaurs

- The dinosaurs reflect that the machines were not intended to serve our present day needs.
- They were constructed to fill the needs for people in the past.
- Why did the Dutch wanted to compute.
    - For aeronautics.
    - For land reclamations.
        - Lorentz and Lely were responsible for the build of the afsluitdijk and the creation of the ijsselmeer.
            - They did the calculation work and had it performed by human computers.
        - It was as a calculational effort a typical Dutch effort.
- Johan van Veen joined Rijkswaterstaat after being in the process control and build analogue machines to solve problems in water.
    - He used these machines as computers in the pre-world war II meaning of the world.
    - This tradition existed until 1972 of this process control.
    - They were not kind of that programmable.
        - So they were not super close to an actual computer.
    - What we today assume as a computer is something that has a screen.
- After the war several people set them self to work to do calculations.
    - Aad van Wijngaarden became director of the computing department of the mathematical centre in 1947 in the Netherlands.
        - He set people to work on building a computer.
- Jan van der Corput on computers in 1952:

Nobody doubts the innocence of automatic calculators, but, like any piece of technology, it is constructed for a specific purpose, and once in existence it can be used for good and for evil. You can't wish them away. Although these machines were devised for destruction, it is now our duty to put them to good use. They might lead to the loss of labour, with all the horrible social consequences that go with that. But they might also help to prevent disaster in the future.

    - He wanted that computers contributed to his ideals.
    - In 1952 the ARRA was launched which he presented.

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            - He showed a program to generate a random number, which was quite something in that time.
            - He built a machine that could compete against the US.
                - Which was an example of Dutch greatness which was great for the press and people.
            - This machine went silent.
                - So this was uncomfortable for other people watching.
            - But the machine never functioned afterwards.
                - After they build the ARRA II which functioned well and was put into use.
- From the ARRA came the ARMAC in 1956 and after that came several other machines.
- The number of jobs all over the world done by computers earlier on was relatively low.
    - Because programming took a lot of effort.
- In Delft there was a group of mathematicians who were trying to build a calculator which would calculate for them.
    - Willem van der Poel was one of them who was interested in a machine that could calculate different constellations of optical lenses.
    - He built a machine in 1950 that was called the ARCO.
        - It was slow, but accurate and it did it calculations overnight.
        - The programs were hard-wired.
            - It was not very easy to put in a new program.
            - But it did the calculations by itself.
    - To profit the telephone traffic, Van der Poel designed the ZERO in 1951.
        - The design was satisfying the need for the telephone company who wanted to have a machine that could supplant the operator in operating rooms.
            - They used these in a setting from 1953 onwards under the name of PTERA.
                - Concerned the first commercial machine that was produced.
    - At the beginning of 1955, Van der Poel designed ZEBRA.
        - In production in 1957 by Standard telephone and cable company (STANTEC).
- In Eindhoven, they built PETER which was an electronic calculator.
    - But Philips which built it did not start mass producing it.
        - Because they had an agreement with IBM.
            - IBM would produce computers and they would buy Philips their parts to build these.
            - And the understanding that Philips would not go into competition.
            - Philips also declined to build the ZEBRA for Van der Poel because of this agreement.
    - They did do a lot of computing and make computers.
        - But they discovered that it was very difficult.
    - They build PASCAL and STEVIN to supplant the PETER.
- All companies tried to build a machines that could function.

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        - Whether it could be faster to wire and calculate than a team of human computers.
        - And for fields were human mistakes were not permitted.
        - They wanted to make these machines a part of national pride.
- All these people above got involved at each other after the war.
    - Process control, science and administration came together.
    - They began projects together.

1984

- European tradition of dystopian literature.
    - Here dystopian futures were present.
    - Orwell would describe a world of systems thinking.
        - That was where he was afraid of.
    - New technology could yield a shift in power.
    - He was concerned of totalitarianism, nationalism, surveillance, censorship.
- The public was also informed of these things.
    - All cold war projects were costly and had to be sold to the general public.
    - Documentaries on nuclear power, space travel etc.
- The rise of computing technology was part of modernity.
- People would begin make movies about possibilities of effects of 1950s computing.
- People were beginning to put faith in the future (dystopian kind of faith) which became dominated by space travel and computers.
- There was public attention on radio, newspaper and television.
    - Many people were aware of the present possibilities of computing.

Change in movie culture, 1970 - 2000

- They went digital:
    - Ment a stage of storage.
        - From celluloid, to DVD, to cloud.
        - It changed the way people watched movies.
            - People now could watch movies at home because their more compact size.
            - Rental stores came up when DVDs came up.
                - These became obsolete when cloud came up.
        - Movie culture was affected by digital culture.
- Themes of movies changed.
    - Themes are influenced by the culture in which the movie is made.
    - Themes like a person falling in love with it's operating system was unthinkable in the 1980s.
- Special effects changed.
    - In the 1970s and 80s they used stop motion and puppets.
    - Then they became computer animations.
- There was a change in stunts.
    - They went from physical stunts to green screen acting.

Week 3, college 1

Computing sounds

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- Philips had in 1961 a small disc which contained the sounds of the PASCAL calculator.
    - But the PASCAL itself never made any sound.
    - Yet sound was extracted.
- Computing sounds have been around from the existing of computers.
- When the EDSAC was build, in 1949 Christopher Strachey send a punched card which hummed “god save the queen”.
    - You could have sound like clicking of the vacuum tubes and other sounds of the computer.
    - Similarly the MIRACLE sang the Dutch national anthem.
- It was even one of the ways to check if a machine was working.
    - Old machines made noise.
    - Helped checking how the machine did.
- The ARRA II went silent because it has transistors which didn’t make sound.
    - The ARRA II in 1954 had speakers.
- The UNIVAC had a stall speaker button.
    - Which meant that by default the speaker was on.
- Sounds were used for:
    - Announcing the program had terminated
    - Auditive monitoring
    - Specific navigation within a program.
        - Mathematicians knew were the program was at.
    - Part of the instruction
        - Program is now here.
- All these sounds were an external control on the execution.
- Computer sounds were used in the mid 60s also the other way around.
    - For music and games for example.

Use of metaphor

- Metaphors were used everywhere in the rising world of computing.
    - Programming was one of these metaphor.
        - Preparing a program, and this as a verb.
        - This as a verb was in 1948
- Metaphors were also used to getting used to a machine.
    - Metaphors are a way of appropriating a machine.
    - Appropriating is doing something with a machine which it isn’t supposed to do.
        - To a bit show off and see how and if it works.
- Calling a wired program what they did earlier was something else than how we program now.
- Use of metaphor:
    - Plugging, assembling
        - Plugging because you plugged things into a machine.
    - Pseudo coding
    - Assembling; assembly routine
        - Assembling became an assembling routine.
    - Autocoding, autocoding systems
        - Autocoding systems to note a difference between what you put in and what the machine did and put together.

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- 1960: programming language compiler became a metaphor.
    - A language that you put into a computer and the computer translated into the wiring.
        - Reassuring names for new things.
- Memory was a new metaphor.
    - It was a way to store information for a while.
- The use of metaphor is by the cognitive psychology explained by the urge that people feel to get acquainted by new things they encounter in life.
    - New things have to be denoted in terms of familiar things.
    - That's the idea of all these metaphors.

IBM and European dwarves

- There were European computer manufacturers that could have been as successful as the American ones.
    - But they weren't
- IBM came out of the crisis in the 1920s as the biggest company.
    - In Europe some of these companies that could compete with IBM existed before WWII.
    - But in WWII they disappeared because of the war.
- One of the main European manufacturers was ODhner in St Petersburg (1873).
    - Was supplying machines all over Europe.
    - But St Petersburg was of course not the place to be in 1917 because WWI and the revolution.
        - Odhner was no longer a private company because of the revolution and closed down.
    - After 1917 you saw clones of what Odhner made in Europe after the war.
        - But disappeared because it found itself in the wrong time at the wrong time.
        - Could have competed on large scale with IBM.
- Sweden: you had Atvidaberg Industrie founded in 1920.
    - Produced a very successful line of calculating machines.
    - In 1970 the company went bankrupt and was sold to Electrolux.
    - But until then it was one of the companies that in Europe functioned on a level like IBM.
    - But after WWII it failed to innovate and did not connect to the trends of electronics and to connect to other machines.
        - It stuck to one machine that was successful.
- In Germany there was Konrad Zuse (1910 - 1995)
    - It produced automating machines of various kinds in competition with IBM.
    - When after WWII the Americans roamed Germany for scientist and technology for their research they skipped Konrad Zuse.
        - They did not realise the importance of electronic computation yet.
        - They thought it had no future.
    - After the war Zuse started building its company further and produced the Z4 in 1949.
        - It was a kind of a computer that was partly mechanical and partly programmable.
        - Was efficient for engineering computations.

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    - In 1967 it was sold to Siemens.
    - But Zuse was successful in the 1950s and late 40s as a computer manufacturer and obtaining various contracts from the German government.
        - The Germans wanted an all German computer manufacturer.
- In Britain it was Ferranti who started commercialising computers.
    - Ferranti commercialised the Manchester Baby.
    - Was relatively successful and would remain in service until the 80s.
- France had Bull in 1930 which sold office machinery like IBM.
    - After the war it went into electronic computers.
        - Like the Gamma 3 and Gamma 60.
        - Were electronic computers which were designed to serve in administration businesses.
            - For automating the office so to speak.
    - The French government supported Bull by stimulating French companies to buy from Bull.
    - It competed with IBM on behalf of the French government.
- In 1962 it was becoming increasingly hard to stay in competition with IBM.
    - The French government could no longer offer support.
    - Some capital was provided by GE.
        - This took away a bit the Frenchness.
    - In 1968 Honeywell took over the company.
    - Honeywell and Bull became a rather successful company afterwards.
        - Within Europe.
    - A US company taking over a European company showed that American dwarfs saw potential in this European company.
        - And also in the market.
    - Also the French word Ordinateur meant something else than computer.
        - It meant to order things.
        - Showed well how the French preferred their way of this thinking than the one of the US.
- Electrologica (1956)
    - A Dutch European dwarf.
    - A spin off from the mathematical centre.
    - They produced a “real” Dutch computer the X1 (1959) and it was supported by the Dutch government.
        - The X1 was in competition with several machines of IBM.
        - Most in the market of science but also in administration.
            - In the administration it had to really compete with the IBM’s sales force.
            - The company underestimated that power.
                - The machine was great, but was much more aimed at scientific calculations.
    - But this machine was really displayed as the Dutch computer.
    - In 1964 they moved to Rijswijk which was proudly opened by the prince at that time.
        - You saw some political push to move into the computer business.
    - The government bought many machines from Electrologica which motivated other Dutch companies to go into this business as well.

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    - Also in 1964 there was noticed that it failed to move into the administration business which was really the largest client.
    - Electrologica's X8 (1964) should resolved this partially.
        - Was expensive in R&D terms.
        - It could speed up things and had great functionality, but too expensive.
        - The success of it was quite disappointing.
    - Electrologica was bought by Philips eventually in 1966.
        - Not in 1964 because it would bring them in competition with IBM were they had a deal with.
- There is one positive exception which is the Danish Regnecentralen (1955).
    - Had a quite successful range of computers that it manufactured.
        - And were sold with stocks.
        - If you bought a machine you would also buy some of its stocks.
            - Therefore if you would buy another machine you would increase your stocks which was positive.
        - This kept it in business until it was taken over in 1989 by the ICL.
    - They also had trained coders so you could have software adapted to your specific needs.
- ICL (international computers limited (1968)) was a British merger of several smaller companies.
    - Were reasonable successful selling IBM compatibles.
        - Which was one of the ways of competing with IBM by selling these cheaper.
    - They are still in business as Fujitsu which is a Japanese company that was founded to commercialise Japanese computers.
        - With this overtake they moved into the European computer business.
- In Eastern Europe the best documented Eastern-European countries was Czechoslovakia.
    - Their there was a big project by Antonin Svoboda.
        - Had been working by MIT during the war and encountered with several important scientist like Bush and Aiken.
    - In Prague he found a laboratory of information processing machines.
        - Part of mathematics department.
    - He started building a machine of it's own.
    - As a communist country it had different rules than Western Europe countries.
        - The machines never commercialised.
        - It was a state organized company that would build machines mainly for the government.
    - In 1951 they designed SAPO.
        - A machine that could do huge calculations in limited time.
        - It was enthusiastically received.
        - In 1957 it was firstly produced.
            - But the factory they produced it was not a great success because he was also working on another machine.
            - This was called the EPOS.
                - Was build from transistors.

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    - This process of the build of computers was stimulated by the government but also slowed down by the government because of Svoboda's connections with MIT.
        - Some of the people in the government didn't trust him.
        - So massive funds came from the government which could speed up the process.
            - But also slow down because of personal agendas within the government.
    - One of the ways in which Svoboda and his fellow engineers got what they wanted was through connections with Western-European countries.
        - They could visit the US and UK.
    - Czechoslovakia became part of the COMECON (1949)
        - All countries under Russian rule had to be.
        - They stayed out until quite late.
        - In 1968 it produced the EC which was rather an IBM clone.
            - Was one of the relative successes of COMECON.
    - One of the process control aspects was Zdenek Trnka who build huge analogue machines to help in steel production.
- In general:
    - IBM and the 7 dwarfs was not really a reality.
        - European countries did play a part until the 1980s.
    - European countries were not too late in taking part in new developments.
        - The big companies that could compete with IBM started late.
        - And had often a agreement with IBM not to compete.
    - All these companies were founded by governments.
        - Governments used partly money from the Marshall plan to get these companies moving and being able to buy "their own" products.
    - Rebuilding the economy after the war was an advantage to these companies in most cases.
        - Also scientist were available in much numbers.
    - What did work against was that most of them were interested in scientific machines.
        - Not noticing that administration had different needs for computing.
        - Big money was in administration business, but they were too late realising.
        - Companies that were in the administration business did not step in because of agreements with IBM.
        - They did not take the administration business serious enough to become successful.
        - Except Bull that focussed on administration from the beginning.
    - In 1965 European governments had to withdrew their fundings.
        - Most of the companies stepped out.
    - Labour costs relatively high.
        - Therefore the costs for R&D relatively high as well.
- IBM was dominating of the market in office equipment in Europe and the US.
    - Also the sales strategy of IBM dominated with the FUD (fear, uncertainty and doubt).

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- With the launch of the IBM System/360 companies were forced a bit to build IBM compatibles.

Week 4, college 1

Real time computing

- Came from several ideals.
- It came through Europe via banks.
    - The banking system was the first initiator of real time computing.
    - Barclays at London had the robot cashier.
    - There was a need for an automated system that noted that cash had been dispend to a customer.
- There was no real need for real time machines.
    - No one needed real time response.
    - That's why it took so long to develop.
        - Jobs that real time computers would need worked already well and fast.
    - Process control already had such machines in form of analogue machines.
- Most people also didn't trust it.
- When the machines were there, companies wanted to sell them to make profit of their investment.
    - But no one was interested except the banks who were the first.
    - But originally it was an American idea which came first in banks in Europe.
- The idea of the cashless society rose.
    - But they thought it was best if the credit card did not belong to a kind of club.
        - But to the customers of the bank.
    - It was useful to have a direct response so the customer would see if he still had money.
- Since 69 you had ATM and in 74 you had VISA who were developing real time machines.
- Barclays had batch jobs at the computer center running since 1961.
    - Then they had the robot cassier installed.
- Barclays needed a central computer which was connected to several offices and were info about bank accounts would be stored and distributed.
    - Burroughs was very active in real time computers and produced the D825 in 1965.
    - After that they started the commercial versions B5500 and TC500.
    - They presented itself as the company of the future and bringing business automation to far-sighted companies.
    - Eventually Barclays and Burroughs went together.
        - Burroughs had a factory in the UK so that machines could be produced.
    - In 1965 they announced the TC500 and B8500.
        - Was announced to be ready in 1969.
- Burroughs was successful of selling their machines to Barclays because of their marketing strategy.
    - But the program's were delayed in production which Barclays was not happy of.

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        - So they installed 2 B6500's.
    - End 1968, Barclays merged with Martins.
- But in 1969 the production of B8500 was cancelled.
    - Burroughs could not afford the production anymore.
- There were also problems with the computer center location.
    - There were trains running by which vibrations had to be damped.
    - Barclays became nervous and brought the machines in batch mode to look if it functioned.
- As this was a pretty slow progression, it were the banks still that were the first to use and test these machines.
- Another example is the universal product code.
    - It was a political feat.
        - There was a necessity for larger firms.
        - The whole process went faster.
    - For smaller companies it became difficult to enter the food market.
- The European part of the story was different.
    - They opted not to join the universal code, but to make their own code.
        - The international product code.
    - Was negotiated in 1973 and 1974 and adopted by 12 countries in Europe.
    - It had some differences with the American one.
        - It had 2 more digits for example.
    - This was one of the cold war frictions were the differences between Europe and US came forward.
        - There were political differences.
    - One of the projects that bindend the countries of Europe was this project in Europe.
        - But it was also a bit sticking up a finger to the US.
    - Overall it was a good example of the use of American tech in Europe to better it.

Agenda's

- Was what people wanted to achieve.
- These agenda's would be about selling machines or an academic discipline.
- Within the academic world there were different agenda's
    - Like Europe was very busy with airplanes etc, the Americans were very busy with their agenda's.
        - Primarily on military and atomics.
- Locally also academics could have different agenda's.
- After the war logic become studied were Turing and Church already began with in the war.
- The first computer would be claimed by anyone who was busy with these kinds of technology.
    - No one was trying to build a computer in our sense of the word.
    - They were trying to solve problems they occurred.
    - Why they were called computers was by us.
- Like Dijkstra was not set on the real time machine.
    - But rather on a large calculation machine that would respond in a large batch job.

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        - This was his agenda.
    - He wanted the cause of computer science determined.
- Also other people had different agendas which sometimes led to projects going 2 ways.

Programs and software

- From the wiring and the tape, you would not be able to spot what was going on inside the program.
    - Feedback came from the audio of the machine.
- Programming in the 1950s is far removed from programming today.
- Wilkes, Wheeler and Gill produced the first programming manual in 1951.
    - Then programs were prepared, because you plugged all certain wires.
        - And assembled.
- What were the alternatives for programming.
    - There was none.
    - It was either plugging or setting switches.
    - Within Whirlwind people were thinking of setting up the program in a more efficient way.
        - They started thinking of autocoding.
- FORTRAN was created in 1957.
    - It resembles the kind of programming we are used of.
    - Started as a flowchart and then it would be transformed into a program.
        - The programming on the punch card could be done automatically.
- The same problem of programming existed also in Europe.
    - They came up with the language ALGOL 60.
        - Satisfied European sense of clarity and order.
    - From US point of view:
        - Academic
        - Not flexible
            - Better in theory than in practice.
        - Difficult to learn.
    - Were came this frustration of Europe from in the programming things.
        - The problem was making existing programs run on a new machine.
            - You would make a program for a specific machine and then a new machine came woops.
            - So either you had to make the program again.
            - But the autocoding systems were about this, the ease of programming.
                - Making it easier to make programs to work on others.
- Who had agenda's in programming?
    - Many
    - IBM and USE (Univac) was one of them.
    - A huge amount of companies which worked on programming and shared their knowledge.
        - They all came with sort of different solutions.
        - Why not pick one language?
            - To some any language would do if it worked and was clear.

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            - like IEEE was good with FORTRAN but was anxious for their community for software to work.
            - IBM would go what ever sold them most.
                - FORTRAN and COBOLD was chosen.
            - IFIP wanted to go as international as possible.
            - SHARE, USe, DUO and CUBE went along with their usual languages.
                - Had no will to go with another.
- The ALGOL conspiracy.
    - Within German cities, logicians, computer scientist etc created a universal language.
    - Which would bind the nations of the world.
    - But a universal language was one of the things that was blown out of proportion.
- A huge of effort of European academics went to ALGOL.
    - Decisions would be made how the language would look and what the structure would be.
- In 1962 IFIP created working group 2.1 which worked on ALGOL.
    - They did not interest in it being the only language because they wanted to reach all people globally.
- ACM wanted ALGOL as standard for publication of scientific algorithms.
    - But not for other algorithms which was their hidden agenda.
- Within the American computer world, ALGOL never made it even as a scientific language.
    - In Europe ALGOL made it as scientific language.
- With programs came software.
    - Beginning 60s software alone did not suffice to uphold a business.
    - But rather a range of computer services.
        - Maintenance
        - Machine building/tuning
        - Batch processing
- There was also no money in software neither in Europe or the US.
- Software companies existed in the 60s, but they did various things.
    - Yes they wrote code, but also built machines and did maintenance like mentioned above.
    - This changed at the end of the 1960s.
- Even the meaning of the word software changed.
    - Delimitation of “software” was vague.
    - Code produced by one of the companies programmers was referred to as code or program.
    - An external company testing a program was viewed as a software project.
    - The meaning what it has today was the unbundling which IBM made.
        - From then in 1968, software became both soft and ware.
            - Soft in the kind that you could “wear it”.
            - Ware in the sentence that it was becoming a good that could be sold.
        - Academics in Europe had some trouble with that idea.

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            - For them software was something extra and not something you did when computing.
        - Even Though a small software industry rose in Europe, it relied on free work for academics who refused to get payed for it.

Software crisis

- Problems with “growing” amount of code.
- Also amount of people working on the code became a problem.
- Code was hardly fault prove.
- In IBM’s OS/360 a real problem was signaled.
    - Within IBM it was solved that it worked for the clients which was wat counted for them.
- In Europe a similar crisis occured.
    - A similar crisis which IBM ran into occurred in banks in Europe.
        - It was less of a software problem, but hardware was the problem which wasn’t there yet.
        - They tried to make it work with computers that were available but not really.
    - If there were problems within companies, companies were often eager to jump in and solve it.
        - Sometimes making the problem worst.
        - But many companies made money by making money for existing problems.
    - There were problems for scientist with getting this new line of business incorporated with existing machines.
        - They had an agenda that was crisis sensitive.
            - Like in the US.
        - Nevertheless there is a difference to the Europe universities and US universities.
            - Uni’s in the US needed a group of people who could start managing and have money to create program’s.
                - If they ran out of money it was their problem.
            - Within Europe uni’s, the government gave money for the program’s.
            - In Europe, computer scientist were mostly working in other disciplines.
                - As a field computer science did not exist.
                - The question was why would they bring it to existence and why bring the cost.
- NATO produced software engineering books with techniques to solve problems.
    - Software engineering as a way of solving a crisis would made a portable way of offering a solution.
    - Software engineering could offer solutions to problems.
        - Even if the problem wasn’t there yet.
- Software engineers could be uni’s and other companies.
    - They all tried to get more grip on the wild growth of computer manufacturers, user groups and all kinds of organizations.
    - The only group that had any experience with the crisis wasn’t there.

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            - Which was IBM.
            - Was not present because their crisis was not experienced as much as a crisis as much as in Europe.
        - All these companies need to agree what software engineering would be and how to solve the crises.
            - Several ways which was suggested were universal programming languages.
                - But was dismissed.
            - Also structured programming was brought forward.
                - Was not dismissed.
                - But the US had other styles for this.
                - Many of the Europeans preferred structured programming which were often mathematicians.
                - The agenda was set by academics.
                - Software engineering was primarily theoretical.
                - The program would not be addressed by one program but by several which were connected.
- Programmer became a profession also in the economical sense.
- Informatics as science, software engineering being an important building block.
- It became useful to program first on paper to prove, and then start to code it.
- At the beginning of the software crisis, Dijkstra really wanted it's way for programming in 1967.
    - It succeeded and had its own idea of what the software crisis was about.
    - But also had its own solution which was structured programming, which worked.

Week 5, college 1

Role of women

- Why are there not a lot of women in history of computing?
    - The subject has been claimed by men for quite some time.
    - Contribution of women became forgotten and unappreciated.
- Many were human computers first.
    - But rose more and more but rather invisible.
    - Because that was the role of women at that time.
- In Europe being a human computer was mainly a women's job.
    - In the US it was exclusively a women job.
    - These were black women.
        - There was a racist profession there.
- In the 80s there was Esther Dyson
    - Was one of the most tech people in the US.
    - She ran a newsletter that was read by almost all heads in computing.
    - She had a fine feeling of what went on in computing.
- Also Saskia Stuiveling was one important women.
    - She fought for transparency in spending of government money and NGO budgets.
    - She came up for open data.

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European valley's

- In Silicon Valley in the US things started boiling.
- What went on there was much more complicated then thought.
- Why were there no European valley's?
    - A valley was really special.
    - All these special projects like the showing of the first pictures and shots of our planet, was something special in the valley.
        - You could call it sort of new hippy culture.
        - Was not present elsewhere in the world at that time.
        - In the 1970s those things happen nowhere but in the valley.
- There were European valley's.
    - But they were more part of utopian city planning.
    - We had academics and high tech companies that were interested in time sharing facilities.
        - Not so much that culture of hippy life acts like in Silicon Valley.
    - That city planning was also some sort of planning of culture in Europe.
    - We had Sophia-Antipolis which was build from scratch and were people would live and find their work at the same place.
        - There were factories, houses, and recreation.
    - This planning did not work out the way that people anticipated.
        - Sophia-Antipolis was one of the most successful.
        - But most failed.
            - People didn't want to live there at all.
            - That was a problem Sophia-Antipol was facing as well.
            - People lived there during the week but went home during the weekends.
            - Thus instead of becoming a active city as was planned, the planning did not always turned out they wanted.
        - This went down with many European valley's
- Twente polytechnic was a very successful valley just outside Enschede.
    - Was well accessible and people could live and work there and go to university.
    - There were huge plans for factories next to the university buildings.
    - But it was never filled up because people did not want to move all the way to the east of the country.
        - When they were already conveniently located in the west.
    - Students would stay in the city of Enschede or go out in the city.
    - The project of Twente failed in which was anticipated.
- It was the interest in the rising youth movement that was missing.
    - It was there in the US.
    - But it was not here in Europe.
    - They were not interested in electronics.
- There was some interest.
    - The Amateur Computer Club by Mike Lord which rose.
    - Groups like this were associated with the amateur computer enthusiasts.
    - They had the ambition to create something new.
        - Were only interested in the seclusive group of their own club.
        - Made them stand out of what happened in Silicon Valley.

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- People wanted to come together and learn together, and even sell together.
    - Was a different way of going about like the hippies in Silicon Valley.
- There are European hippies.
    - Here there was no blend with those tech enthusiast.
    - Computers were no part of it.
    - Computer enthusiasts would come together, but computers were no part of it.

Appropriation

- Appropriation was the reason that the computer became such a machine as it did.
- Appropriating was to assemble the machine yourself which came in a package.
- 1977 was the founding of the Hobby Computer Club
    - Activities for the members
    - Exchanging knowledge and software
    - Programs in the newsletter
    - Radio-broadcasting programs with NOS
    - No political agenda
    - Identity, though!
        - Clear sense of identity.
        - You needed to know your stuff.
- People in the Hobby Computer Club were often professionals.
    - But liked to keep going at home.
- There were many other computer clubs.
- Over the course of the 80s they changed.
    - Becoming customer organisations HCC and PTC mainly:
        - Guarding the interests of users.
- Often these clubs were connected to one type of machine.
- All these clubs had gatherings to share and swap.
    - They exchanged program's with each other.
- Appropriating was not politically motivated in the 70s.
    - But it was becoming politically motivated in the 80s.
    - Most notably in the squatier movement.
        - In France, Germany and the Netherlands squatered a home and took it as their home.
        - They tried to seclude themselves from the ordinary world.
- From this Squatier movement the computer was received.
    - Was used for political purposes.
    - They thought government might do surveillance through machines.
    - They used computers for a few reasons.
        - Some tried to show that computers were anti-democratic.
            - And were a computer club without any structures and how to better ourselves with computers.
        - For example few hacked a bank and returned the token money and showed how and express it to the press.
            - First they were heroes, then hacking was seen as a crime and few years later they were arrested.
- In Europe there was a lack of political activities in the 60s and 70s.
- But there was a rise of computer science in universities.
    - There was continuity.

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    - The computer scientist in Europe were much not interested in those computer clubs, but they respected the technical knowledge of these people.
- In the Netherlands there was no academic setting for information science.
    - It was there in the US, France and in the UK.
    - But here the information scientist were left out of the universities.
    - It is not the case that no administrative machines were used in companies or elsewhere.
- In the Netherlands we had the SSAA.
    - It was a study group because there was no academic place for them together.
    - There were connections with academic computer sciences.
    - They had a journal: informatie.
        - Providing information, courses and seminars.
    - They were a counterpart between the hobby computer clubs.
        - They almost had nothing in common.
    - One of the members was Max Euwe.
        - UNIVAC ambassador, advisor in automation.
        - He could play chess and was matician.
        - 1964: professor in automatic data processing.
        - He wanted to reach more people than just members of the SSAA.
    - People of the SSAA were often more serious users.
    - They build software like VISICALC for also serious companies, but also hobbyist.
        - People who worked in offices and liked to go on at home.
        - So there was a overlap with the computer hobbyist.
- Funded by the Dutch government, Teleac television courses about microprocessors and programming.
    - All people in the Netherlands could do this course.
    - Bringing the computer to the public funded by the government was typical in the west.
    - Bringing the computer to the people in this way was another way of appropriation.
    - People started doing things on computers that it wasn't made for.
- Christ Titulaer promoted technology and science.
    - He was director of the television academy.

Computers in education

- Displaying pictures on learning books showed the possibilities that these machines could have on society.
- Within various textbooks pictures were present of the computers.
- Subjects in textbooks were:
    - Binary arithmetic
        - They thought of it as the language of computing
    - Flow chart
    - Even some programming
        - Could not take place because the classrooms had of course no computers.
        - What did happens was that teachers had connections to one of the computer centers.

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            - Students their programs would be taken to the computer centers and next week their results would be back.
- Lifting with the hype of these machines was were these subjects were all about.
- One of the problems of 1950s education was trying to evoke learning processes.
    - After and before the war teachers realised that it was not that easy to make students think by themselves.
    - Having them to think why things happen instead of blindly doing equations.
    - Learning abilities: finding out how it works.
        - Was a hot topic that time in education.
    - In general people wanted more learning abilities.
        - Because they thought these future students would not have the nice possibilities of their parents.
        - Also cold war was going on so they had to compete with Soviets.
    - Computers in education could contribute something.
        - They could adjust to students needs.
        - But they also could help the teacher out.
    - Several computer companies set out to find solutions to education problems.
    - The connection between computing and education was mostly at cultural level.
        - Thinking about education as it was a technical problem.
- Several machines were built that could take multiple choice quizzes.
- Programming instruction was some sort of a machine with multiple questions but in a book form.
    - Books were much cheaper than machines.
    - Students could make a question, and with a slide reveal the answer.
    - Even without computers being present actually, the trust in computing technology rose.
- Computers in education would become common in the 1980s.
    - Computing companies helped to facilitate computing rooms.

Week 6, college 1

Difference computer club US and EU

- In the US their computer clubs there was this blend of hippy culture and academics.
    - This was not the case in Europe.
        - This hippy culture was not there.
    - Just like the Valley's .
- Also within the Homebrew computer club was paying for software which became normal.
    - In the Hobby computer club this was not the case.
- So like the Valley's, the blend of 2 cultures did happen in Europe.

Valley's

- Making people to move and work in the valley's, having tech companies to have factories there was not that easy in Europe.

More appropriation

- In the 1980s and 90s, people experienced a European sense of community.

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    - Was in many ways opposed to US culture.
    - The squatter movement show this.
    - There were many groups in society that did not want to associate to the US.
- George Orwell 1984 was in Europe read as some sort of opposing society like the US which with its technology could make this happen.
    - In US this was more read as a cold war book, thus that the Soviet Union could do such a thing as happens in 1984.
    - Apple made a commercial to try to break through the European vision of this book.
        - It was in Europe very welcomely received, also in the squatter movement.
- The squatter movement in the 1980s was very European.
    - Hardly any squatter movement in the US.
    - It was politically active in the EU.
        - Trying to find a anarchistic way out of the grown political situation.
    - This may be compared to the hobby computer clubs transit to talk about software prices.
    - The squatter movement gave affordable ways of building your own movement.
        - But added the concept of hacking.
        - They saw hacking as a political statement.
            - So they made it a political statement with their hacks.
            - These hacks were publically displayed.
        - The whole idea that hacking was a computer crime was when Wernery was arrested in 1988 for computer crimes.
    - But the computer clubs also brought appropriation.
        - Hacking was of course also a sort of appropriation.
            - Something they done with a computer that it wasn't supposed to do.
        - One of the most typical European cultural expressions in computing that originated from the hacker movements are the Demoscenes.
            - These demo scenes were not restricted to the squatter culture, but originated there.
        - People started tinkering with computers and made art also with it.
            - Became fashionable in the squatting culture in the 1980s.
            - Things like graffiti which you would find on words, appeared on computer screens.
            - These arts would be typically displayed in diskmags (1984).
                - These were magazines like that were sent around the world for people who were interested.
                - Contained arts, discussions etc.
                - They borrowed their structure from typical magazine structures.
                - This went online that sharing and the computer was the distributing factor.
            - The demoscene did not take place in the US until the 1990s.
        - Another form of appropriation was not taking place in the squatter movement.

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            - Originated from squatter movement but also from academics.
            - This was the idea of a free software movement .
            - Rose from the US, MIT specifically in 1983.
            - They insisted on software being free to use.
                - Also free to alter and understand.
            - It was one of the things that was picked up by European movements.
                - Both by academics and squatter culture.
                - But they were thinking in terms of free software in the sense that you don't have to pay for it.
            - The open source movement that originated from this free software movement were meeting the idea that originally happens.
                - Happened in 1998.
        - Another form of appropriation was sort of forced to people by European governments.
            - The home computer projects (from 1986 onwards) made owning a computer at home not that costly.
            - Mostly people would play games on these computers.
                - This was not what the government had intended when they started these projects.
        - More appropriation was taking place with the invention of the internet.
            - It brought several events that were spreading over countries.
            - Something like the project of the digital city in the Netherlands.
                - It was the place where you would go to the pub virtually for example.
                - You could chat and communicate.
            - This appropriation of the early internet was extremely popular in Amsterdam.
                - People wanted to be part of this digital city.
                - In hacking culture these kinds of projects were very popular.
                    - It was a new way of using a computer.

Rise of internet (Europe)

- As in the US, in Europe the internet was a consequence of free desires.
- The wish for this world wide encyclopedia was both a wish in the EU and US.
- The desire for a full torent military network came from NATO (eu).
- The desire to connect the various networks was also a European wish.
    - Was an easier wish in a sense.
    - And even a politically funded wish.
- An example is the French Minitel.
    - From 1978, France Telecom promoted this machine.
    - It was a way of advertising telephone connection.
    - The minitel was using the telephone network.
        - And French telecom which was a state company would try to sell more telephone connections by Minitel.

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    - The culture in France was that people would have a central phone on the village square, which all people would use.
        - French telecom had the infrastructure for everyone to have phones at home, but not really many people got one.
    - Minitel was a success.
        - Not only because of the machine, but also because of the content of the network.
        - You could make use of info from various municipalities and government institutions.
            - All of them offered content that could be viewed on the Minitel.
- Electronic mail was one of the reasons for success (1981).
    - Many companies, universities and government institutions would have electronic mail from the beginning or within 3 years from the origin.
        - Not many people at home would have mail addresses in the beginning.
        - It was as popular in the EU as in the US.
    - Because French Telecom was a national company, it was distributed across the whole country easily.
        - There was no problem with making connections between various companies.
- In the Netherlands we had Viditel (1980) which was an initiative by the Dutch telephone company.
    - In an attempt to sell their network projects.
    - The Dutch had a state owned telephone company and their network was used to distribute information.
    - But Viditel had a lack of content, opposing France.
        - They were hoping it would evolve itself, but it hardly did.
        - The price was also very expensive.
            - The ordinary citizen could not afford it.
        - They also had competition from the television and radio station which were also state owned.
            - They launched teletekst which was not extremely rich in content, but it came with the TV (1980).
    - But the Dutch telephone network had a network established in in 1978 using circuit switching and designated telex lines.
        - That was called Datanet 1.
        - After the starting face they went for a package switching protocol that would allow for more traffic on the lines.
        - So just as France they used a existing system which was already there.
        - But again it was not very affordable.
        - Also it was a very slow connection for the price.
        - But it was kept in use for quite long.
            - It was pretty successful for people who wanted to use it.
            - Like universities, banks, big companies.
                - They could transfer info quickly and to the right location.

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            - It was for them an affordable and better alternative to the phone.
                - Could directly be sent to another computer.
                - Was less faulty.
        - In the early 1980s it was adopted by the Municipal Administrative services.
            - Established a financial base.
        - Until the end of the 1990s was used for network payments until the end of the 2000s.
            - Then people began using the internet.
- So these 2 stories show that national network services could be an advantage and a disadvantage to these networks.
    - Could be a disadvantage when people would not push info on it.

Game industry

- Playing games was a way of appropriating in the 1950s.
- Alan Turing found a chess algorithm in 1952.
    - But this could not be played at the machines of that time.
    - But he proved that there was an algorithm for chess.
    - But it was a way of tinkering at first.
        - Showing of to colleagues if you could program it.
        - First time this tinkering was a goal in the 1960s.
            - This was Nimbus which was a small game.
    - And they wanted to program a game that could be an opponent to a human being.
        - And could show what a computer could do.
        - This was also the idea with the game Spacewar (1970).
            - It was mostly played at computer at university.
            - You would even have tournaments.
- In the early 1970s, various games would be used to make large arcade hall machines.
- At the same time in universities there were made text based games.
    - These kinds of text adventure games were present in many numbers.
- The real rise of the industry was in 1980 with the launch of Pacman.
    - This was sold as software.
    - This was a different concept that having a whole machine to play a game.
    - This whole idea of games being sold as software both launched the game industry and also got it into its first dip.
        - Because some companies producing bad games.
        - But from this dip it rose quite fast and focussed on multiplayer, graphics and various other options.
- Video games would also be used for educational purposes.

Rise of computer science and information science

- The official jobs that rose and educational purposes around it.
- This professionalization took place in the 1960s, 70s and 80s.
    - From there continued to this day.

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- In France there the rise of computer science and building of machines and academic discipline came from within the departments of management (social) sciences or engineering schools.
    - That's were people were interested in when building these machines.
    - Management sciences were very interested in these computers.
        - Most interested in automation.
- Within the France government and academia the use of these machines were welcomed.
- In France founding a new academic discipline required enough students that wanted to be thought that discipline.
    - These disciplines grow organically and stayed within management sciences till the 1970s, and from the 1980s the departments of computer science were formed.
- The computer and information sciences were formed more or less at the same time.
    - Starting in the 1960s within the management schools and became schools of their own in the 1980s (in France!!!).
- In the Netherlands things went somewhat differently.
    - Adding a new discipline required students and money, but also convincing the government.
        - They fund new fields of studies.
    - Because of this computer sciences rose rather late.
    - Within the mathematics departments computer scientist were active.
        - But actual bachelors date from 1981.
    - Information science was even later, in 1984 in the Tilburg Catholic School.
        - Which at the time wasn't even a university.
    - It was quite late because of mathematicians that claimed the subject.
- In Belgium and the UK information science is mostly part of the department of management sciences.
    - Like in France.
    - Lifestyle informatics would be part of the medicine department.
- In the Netherlands the founding of bachelor courses took quite a while.
    - But the national and international organisations were already there.
    - Peer reviewed journals came quite late, in the 1970s.
        - People would publish early in mathematical journals.
    - The rise of Jargon.
        - Using metaphors, and using ways Dijkstra said you should code for example.
    - Writing history was already there in the 1970s.
        - When people start writing their history, you could say people start their own profession.
        - This is an indication that the field is becoming professionalized.
    - But the founding of the bachelor courses was much later than in other EU countries.
        - That reflects the academic system here.
        - You could study that contained a lot of computer or information science if you picked the right courses.
- In France and the US there are also many differences.

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    - In France study courses you knew before you would start which courses you had to take.
    - In the US there was a much more open structure for courses.
- Within the Netherlands we moved from the French system to the US system in the 1990s.
- Professionalizing computer science happend much earlier than information science in the Netherlands.
    - In the late 1980s that computer science departments could exist separate from mathematics departments.
    - Only in the last decades separate AI departments have been evolved.
- These kinds of developments have taken place all over the world in various fashion.
    - Information science and computer science rose together in some form.
    - Either as one department or separate departments.
        - More or less simultaneously.
        - Netherlands are an exception.
- It has been suggested that the 3 traditions (administration, process control, science) are in a way connected to the information sciences, AI and computer science.

- Administration was definitely interested in information science.
- Within the science department the need for large scale calculators set the wish for computer science.
- But keep in mind that after the war these groups came together and really never left each other.
    - The overlap between these traditions melted after the war.
    - People needed each other.

Week 7, college 1

- For a decade our culture had been described as a digital culture.
    - Has been described by our way of digital interaction.
- What makes the world digital today.
    - This digital part of our culture is hyped.
        - We do a lot in a digital way.
        - But all these things have their roots long before we are now.
    - The culture has changed from the decades before.
        - Some of the ways of doing are connected to the fact that for example we all have a phone.
        - Culture has become more wired with the digital world.
        - Also everyone almost has accounts on several services like gmail and facebook.
            - That takes large or small parts of our lives.
            - It's part of our society, but in a virtual reality.

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        - In a game you take part in a society that is not our real society.
- Several things in the digital culture relate back on the non-digital culture.
    - Like the mail icon being an envelope that is hardly used now.
    - The envelope represents a part of non digital culture that will slowly be not recognized by people who have not grown up in the non digital culture.
- For example e-learning can be the future which contributes to this digital culture.
    - There are various groups in society that hold an agenda that is fitted to e-learning.
        - Especially governments for who it's cheaper to have more e-learning.
        - The agenda of teachers of course opposes this.
        - Also a student has more freedom with e-learning, but also completely loses track because of a lack of structure and this freedom.
    - Did education change by making it digital?
        - Definitely e-learning in de form of canvas and electronic learning environments have changed the outlook of education.
            - And thereby our habits.
            - Also we can also upload sheets, which makes students rely on these sheets.
        - Partly it's digital, but also we changed little things in our behaviour but nothing much has changed completely.
            - What you have to learn and how it is learnt to you is still the same as before.
            - The world has not become easier.
- There are historical theories that describe the change in society, how does these take place.
    - The theory of appropriation.
        - People start tinkering with machines which works and continue to do more with it.
        - You can see several ways of appropriation in education.
    - Stabilisation process is also an option.
        - You describe it as an existing technology that is there.
        - Someone has to be taught to use it and then someone can continue using it.
- Also sociological can describe these changes.
    - Through marketing techniques.
        - Manufacturers push products in society and watch if it becomes a hype.
        - Or the market is asking for a new product and pulls manufacturers to produce it.
    - Both ways above are ways of describing changes in culture.
- Also the other way around can describe this change.
    - Reinforcing existing social tendencies.
        - Like social media reinforcing the existing sort of gossip things.
        - So not something that did not exist before is changed.
    - To some thinkers this change makes our culture not that digital at all.
        - If digital, then blended with our normal culture.

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        - When someone has their needs in the digital world and the real world, these 2 are blended and people do meet their needs in both places.
- Technology has not necessarily a purpose.
    - The makers give it a purpose.
    - But antripolitist think that there's not a real purpose and that appropriation is making use of the technology, not appropriating it.
- Digital culture is for a lack of a better word, and is fitting in some senses.
    - But not in every sense.
- Describing the digital culture was something people did in the 1980s.
    - Words like communities shifted meaning to the digital age.
        - Being at the same spot was not necessary anymore.
    - Like Esther Dyson putting "the real world" in quotes.
        - Because the real world was partly digital.
        - It was just as real as the physical world around us.
- The are 3 general ways of describing that digital culture.
    - The first is the knowledge society.
        - The idea came from the observation of people that had knowledge that took more advantage of the internet then people that had no knowledge.
            - Something that struck many as unusual or even unwanted.
            - It was expected that the internet and the ideal of which it originated from that the knowledge of the whole world available to everybody would have a democratic effect on everybody.
        - Knowledge is still something very important.
            - More important is what to do with the knowledge once you have the knowledge.
        - The next step in the knowledge society is how to use the knowledge and how to assess it.
    - The second is the risk society.
        - Is all about assessing risks in all kinds of situations.
        - Assessing risks and making an educated guess of what will happen.
        - Risk society involve a lot of calculations.
            - You emphasize the calculation efforts that go into our culture.
            - That determines our political path partly at least.
            - The world is run partly on assessments based on models that try to predict future events.
        - Digital aspects are very present.
            - Algorithms, facial recognition etc.
        - Using models to make political decisions is a dangerous habit.
            - It can save lives.
            - It may also harm our democratic rights and virtues.
            - The model is always holding some variables that are not part of reality.
            - It reinforces a less democratic system.
    - Information society is the third one.
        - It has it's history of its own.
            - It precedes the knowledge in a sense that in the 1980s information was still key.

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            - Finding the right information was still important.
        - With risk society you have a part of that information society return again.
            - The information that you base your models on has to be obtained in a suitable way.
            - The information society thus has changed a bit.
    - Political decision making is involved in all of these above.

Look back

- The simple story, people wanted to own a computer, so we started making them is not true.
    - There were different desires, different agendas.
    - People just wanted to make money.
    - People had ideals and fitted other their people their agenda and needs.
    - Unintended use, appropriation.
    - Gaining trust was a thing.
        - Machines first were not trusted.
        - And when they worked people trusted it.
    - Confluences.
    - Creating new markets and creating desires.
    - People had simple desires to get into modernity.
        - Or bluffing your way in.
- Today we find that computer like devices meet every aspect of our society.
    - Thereby changing our perception of society.
        - In the form of information, knowledge and risk.
    - In a economic, cultural, social sense digital culture has changed our outlook.
        - Also in entertainment, politics and understandings/epistemological.
    - In that sense the world has somewhat become digital.
- We do get another outlook on live.
    - For example we get a different outlook on learning and ourselves even.
    - There's always a catch.
        - Some people don't want that other outlook.
            - Some people oppose the use of machines.
            - It's not a disadvantage this catch or an advantage, it simply is there always.
    - What we define as the way how we go about information, that's what defines our culture.
    - The catch can become more serious.
        - Were privacy is involved.
        - Like the COVID19 apps that illustrated how far the digital culture might go and how it undermines typical values of privacy and democracy.
            - But it might save us from death.
    - Keep in mind that agenda's are in play, every time.
    - Judging information can be very complicated.
        - Even simple stories can be complicated.
        - Everything can be found on the internet.
            - That's the problem
        - How does a story get on the internet or removed.

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            - Depends on the stories that you want to tell.
            - What do you see as a fact and when is something a fact.
                - Is decided inside the internet community.
                - If it's not a fact then its put at the side as unlikely.
        - A story can not be as simple as is described.
- Algorithms give rise to several possibilities.
- Also there is an uneven spread to who profits from the digital culture.

Digital divide

- There is a uneven use of internet and computing across the world.
- This uneven distribution is called the digital divide.
    - Is as old as is the existence of the internet.
- There are 2 main definitions.
    - The first one is mainly geographical.
        - Like the pictures of the world and which countries have connections to the internet.
        - This definitions is perhaps less suitable since internet use were there is no connectivity is sometimes not much lower than here.
            - Simply the amount of connections is lower.
            - People might use the internet in groups.
    - The idea of a digital divide has been contested in the last 10 to 15 years.
        - First because the gap seems to disappear.
            - Either in number or the way how internet is used locally.
        - Also because it is perceived as a problem that will disappear in the future.
            - Since technology always had the effect to spread across everyone during the centuries.
            - Television and radio are an example of that.
            - But the divide is still a reality today.
    - Another definition of digital divide has to do with education or purpose.
        - Differences in perception.
        - The simples examples are age dependent.
            - Older people tend to make less use of internet than young people.
        - Also part of educational system the internet is a part of information you need and it can also guide students to information they need.
            - In other educational systems that works far less well.
                - In schools with pupils with less favourable environments are in their education of the internet less inclined to make proper use.
            - The internet makes existing social differences worse.
    - Trying to bridge these digital divides is something that seems to be growing more complicated.
    - Also the second order divide, democratic divide is something that influences it.
        - People from less democratic societies are less teached in using internet.

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